

Xiang Ge (Xiangge) Luo

☎ +41 77 985 76 18 | ✉ luox@ethz.ch | 🌐 xgluo.github.io

Research interests

Probabilistic graphical models, stochastic processes, population genetics, phylogenetics, cancer evolution

Education

ETH Zurich

Basel, Switzerland

PHD IN COMPUTATIONAL BIOLOGY - **ETH MEDAL NOMINEE FOR OUTSTANDING THESIS**

2020.11 - 2025.12

- Thesis: Modeling tumor mutation trees for evolution-guided precision oncology
- Advisor: Prof. Dr. Niko Beerenwinkel | Second advisor: Dr. Jack Kuipers
- Doctoral committee: Prof. Dr. Ben Raphael, Prof. Dr. Jasmine Foo

ETH Zurich

Zurich, Switzerland

MASTER OF SCIENCE IN STATISTICS

2018.09 - 2020.09

- GPA: 5.88/6 – Graduated with distinction
- Thesis: Learning Bayesian networks from ordinal data
- Advisor: Dr. Jack Kuipers

University of Waterloo

Waterloo, Canada

BACHELOR OF MATHEMATICS

2013.09 - 2018.04

- GPA: 92% – Graduated with distinction, Dean's Honours List

Publications

Xiang Ge Luo, Jack Kuipers and Niko Beerenwinkel. Numerical approximations of population size distributions for multi-type branching processes. *Manuscript in preparation*, 2026.

Xiang Ge Luo, Jack Kuipers, Kevin Rupp, Koichi Takahashi and Niko Beerenwinkel. Bayesian inference of fitness landscapes via tree-structured branching processes. *Bioinformatics*, 41:i160-i169, 2025. (**ISMB/ECCB 2025**, **Best Paper Award**, acceptance rate 17%)

Xiang Ge Luo, Jack Kuipers and Niko Beerenwinkel. Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees. *Nature Communications*, 14:3676, 2023. (**RECOMB 2022**, acceptance rate 21.3%; **SIB Remarkable Outputs 2023 Award**)

Xiang Ge Luo, Giusi Moffa and Jack Kuipers. Learning Bayesian networks from ordinal data. *Journal of Machine Learning Research*, 22:1-44, 2021.

Xiangge Luo and Alexander Schied. Nash equilibrium for risk-averse investors in a market impact game with transient price impact. *Market Microstructure and Liquidity*, 5:2050001, 2020.

Research Experience

Computational Biology Group, ETH Zurich

Basel, Switzerland

SCIENTIFIC ASSISTANT

2020.11 - present

- Advisor: Prof. Dr. Niko Beerenwinkel
- Developed mathematical and statistical models to infer cancer evolution from cross-sectional single-cell phylogenies, aiming to predict drug resistance and facilitate evolution-guided precision oncology.

Institute of Molecular Systems Biology, ETH Zurich

Zurich, Switzerland

RESEARCH ASSISTANT

2019.03 - 2019.07

- Advisor: Prof. Dr. Mattia Zampieri
- Analyzed E. coli drug response metabolomics data to investigate regulatory and functional relationships between metabolites.

Seminar for Statistics, ETH Zurich

Zurich, Switzerland

STATISTICS LAB RESEARCHER

2019.03 - 2019.06

- Advisor: Dr. Markus Kalisch
- Analyzed mouse experimental data using mixed models to assess bone loss and identify early osteoporosis markers.

- Advisor: Prof. Dr. Alexander Schied
- Derived the Nash equilibrium for risk-averse investors in a market impact game with transient price impact and quadratic transaction costs over finite and infinite horizons.

Awards & Scholarships

2025	ISMB/ECCB 2025 Ian Lawson Van Toch Memorial Award for Best Student Paper , ISCB	United Kingdom
2025	ISMB/ECCB 2025 Conference Fellowship & Dependent Care Grant , ISCB (USD 950)	United Kingdom
2024	SIB Remarkable Outputs 2023 Award , Swiss Institute of Bioinformatics	Switzerland
2018	Undergraduate Student Research Award , NSERC (CAD 4,500)	Canada
2018	President's Research Award , University of Waterloo (CAD 1,500)	Canada
2016	President's International Experience Award , University of Waterloo (CAD 1,500)	Canada
2014	Isabel Farrar Undergraduate Entrance Scholarship , University of Waterloo (CAD 4,000)	Canada
2014	President's Scholarship of Distinction , University of Waterloo (CAD 2,000)	Canada

Conferences, Seminars & Workshops

2025.07	Contributed talk , ISMB/ECCB 2025 Conference • Bayesian inference of fitness landscapes via tree-structured branching processes	Liverpool, UK
2025.04	Poster , 6th Zurich Precision Oncology Symposium • FiTree: fitness inference from single-cell phylogenies applied to acute myeloid leukemia data	Zurich, Switzerland
2025.03	Invited talk , D-BSSE Departmental Seminar Series, ETH Zurich • Fitness inference from single-cell phylogenies using tree-structured branching processes	Basel, Switzerland
2025.02	Contributed talk , PhyloBasel Seminar Series, ETH Zurich • Bayesian inference of fitness landscapes via tree-structured branching processes	Basel, Switzerland
2024.06	Award talk , SIB Days – The Swiss Bioinformatics Summit • Mining tumor mutation trees with TreeMHN for evolution-guided precision oncology	Virtual
2023.06	Contributed talk , SKINTEGRITY.CH Annual Retreat • Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	Spiez, Switzerland
2022.06	Contributed talk , Mutual Hazard Networks Workshop • Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	Virtual
2022.05	Contributed talk , RECOMB 2022 Conference • Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	San Diego, USA
2022.04	Contributed talk , OLISSIPO Exchange Week • Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	Lisbon, Portugal
2022.03	Poster , Ascona Workshop 2022 • Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	Ascona, Switzerland

Teaching

Spring 2023	Teaching Assistant , Statistical Models in Computational Biology	ETH Zurich
Spring 2022	Head Teaching Assistant , Statistical Models in Computational Biology	ETH Zurich
Spring 2022	Teaching Assistant , LSZGS Advanced Block Course Computational Biology	ETH Zurich
Spring 2021	Teaching Assistant , Statistical Models in Computational Biology	ETH Zurich

Supervision

2025	Marie Dogo , Master Student, Mines Paris - PSL
2023	Laura Quintas , Master Student, INESC-ID (Co-supervised with Monica Baciu-Dragan)
2023	Laurenz Keller , Master Student, ETH Zurich (Co-supervised with Pawel Piotr Czyz)
2022	Jiayi Wang , Master Student, ETH Zurich (Co-supervised with Pawel Piotr Czyz)

Academic Services _____

Journal Reviewer for Genome Research, Bioinformatics

Conference Reviewer for RECOMB, ISMB

Languages and Skills _____

Coding Python, R, C++, MATLAB, Git, Snakemake, bash, $\text{\LaTeX}$

Languages English (working proficiency), Mandarin (native), Cantonese (native)